

Overview of O-CEI concerns and limitations

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What are you expected to do?

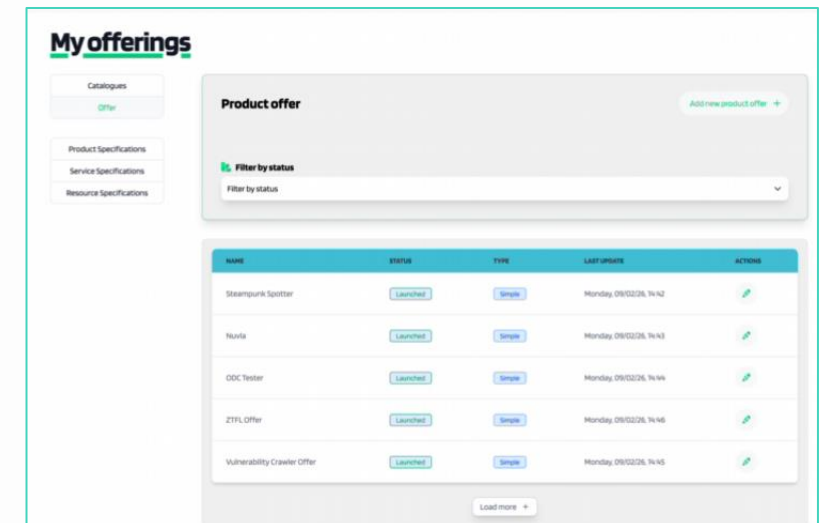
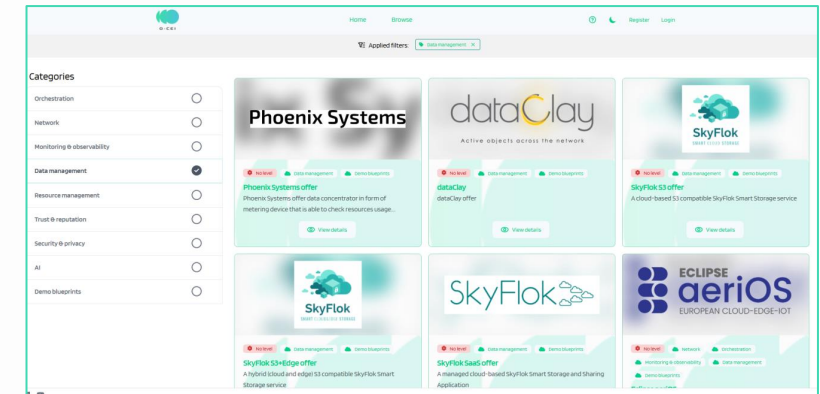
- ❖ Meet your goals, deliver the reports and accomplish the milestones and KPIs as per the **Individual Mentoring Plan**.
- ❖ Provide added value to the pilot, either via direct integration in their premises **or via adjusting with your own infrastructure**.
 - ❖ *"it must be highlighted that final deployment over O-CEI pilots' premises and infrastructure will depend on the current availability, thus not guaranteed. Applicants must plan accordingly"*
- ❖ Upload your solution (dataset, software, tool, methodology...) to the **O-CEI Marketplace**.
 - ❖ Only metadata – description. The source code can remain wherever you prefer.
- ❖ Not necessary to run/deploy via O-CEI platform (to be ready by M21 – September 2026).
- ❖ Not necessary to be part of a blueprint.
 - ❖ However, we will discuss if possible.

O-CEI Key Elements

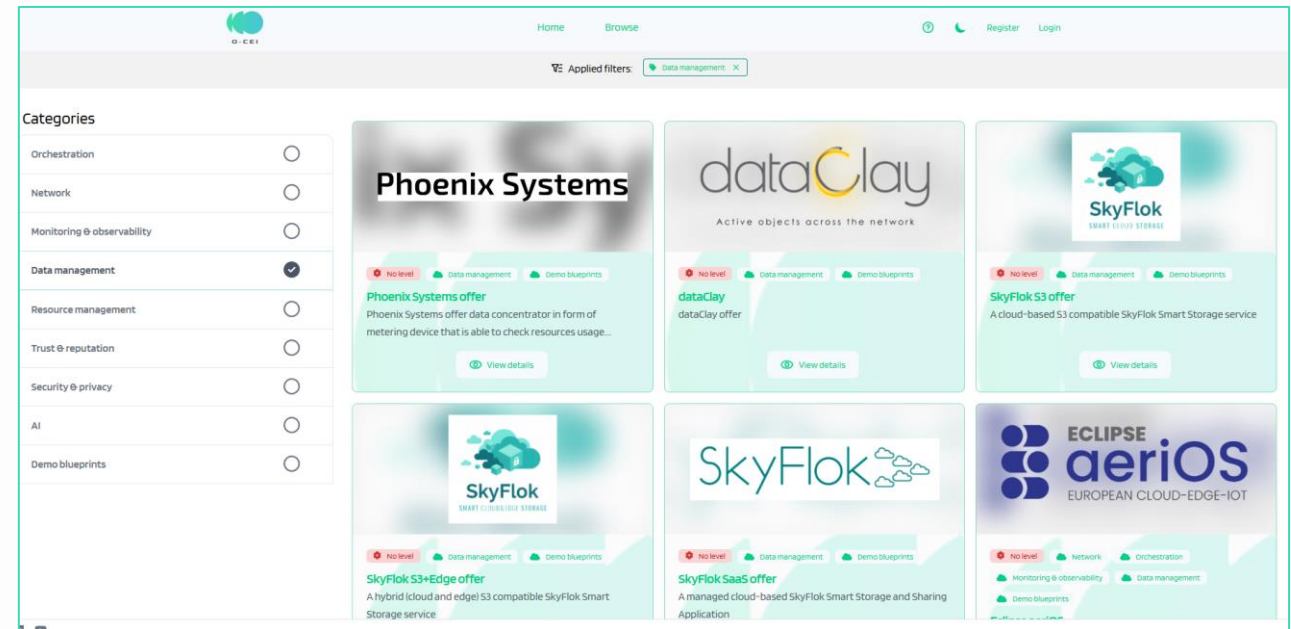
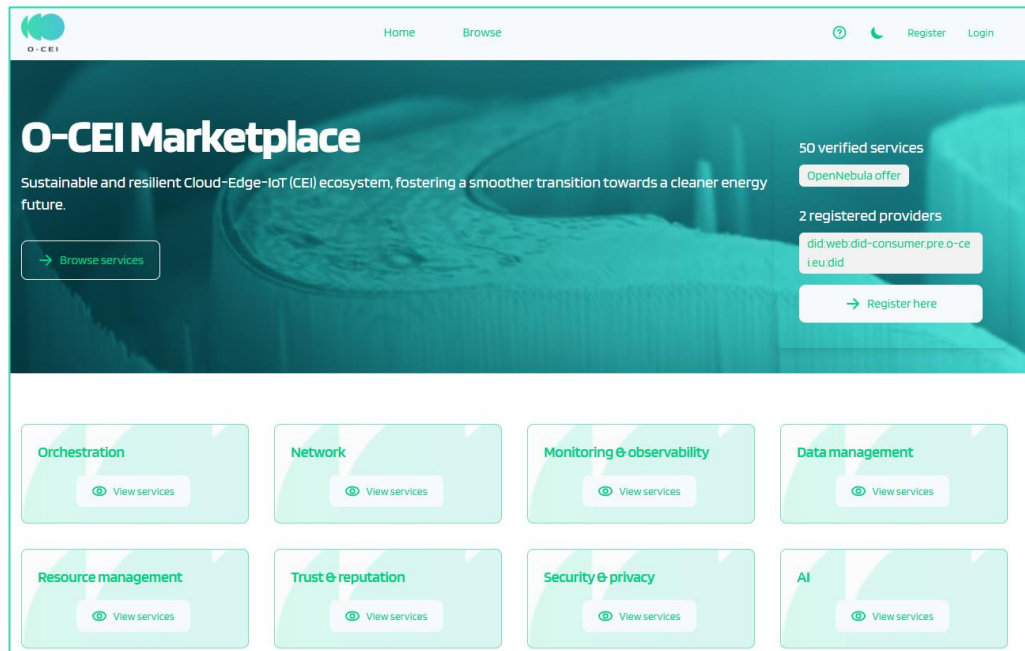
O-CEI orbits into three key elements

O-CEI orbits into certain key concepts:

- **Marketplace:** Offers digital assets that consumers can retrieve, including blueprint products, best practices and guidelines for their deployment.



A **catalogue service** where all digital assets and blueprint products are registered and indexed



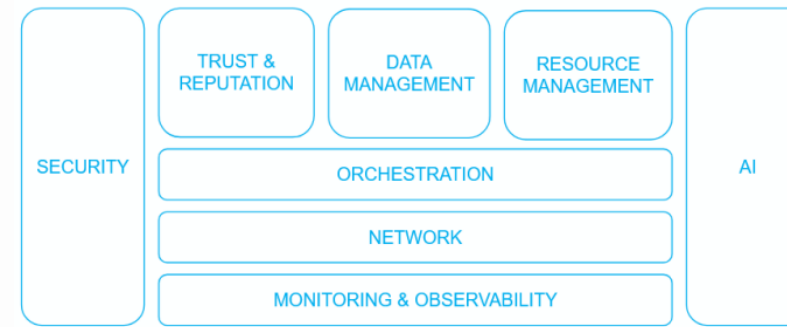
<https://marketplace.pre.o-cei.eu/search>

Digital assets offered in the Marketplace

Already uploaded **more than 40 software utilities (offers)** to the Marketplace:

- **Orchestration:** of services, resources, balancing, scalability, smart allocation...
- **Resource management:** resource registry, discoverability, federation, comms...
- **Network:** Tunnelling, communication/service mesh, access networks, TSN...
- **Monitoring and Observability:** Tracers, logs, alerting, visualisation...
- **Security and Privacy:** Anomaly detection, Smart contracts, Decentralised ID, VC...
- **Trust and reputation:** Distributed ledgers, trust algorithms, auditing...
- **Data management:** Interoperability, sharing, storage, semantics, processing...
- **AI:** Federated Learning, Distributed AI, Agentic AI, Gen AI, Edge/frugal AI, MPC...

Other assets will exist: data sets, AI models, methodologies, best practices...



O-CEI technical partners

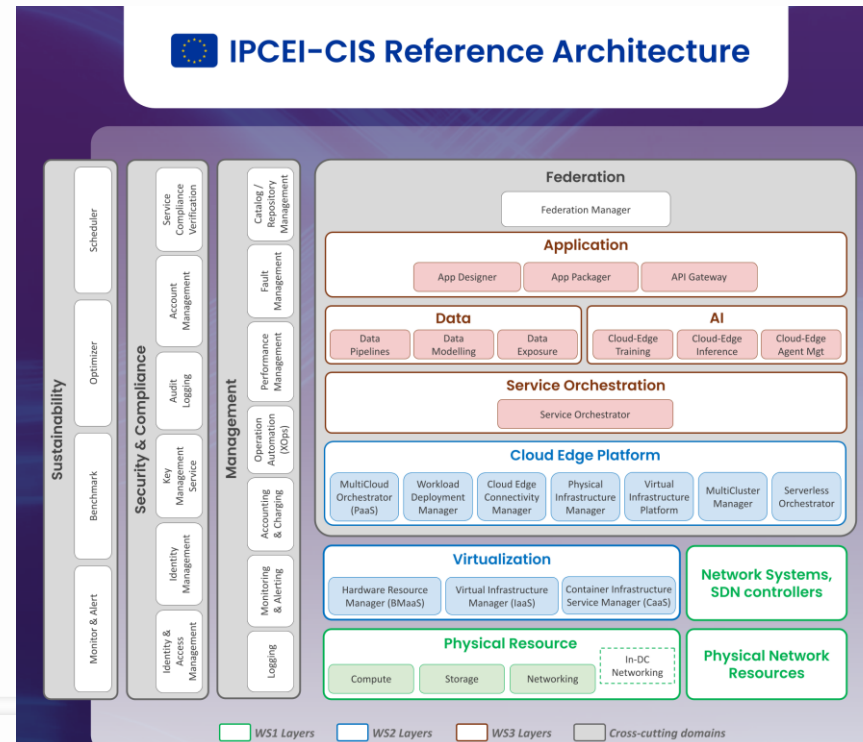
O-CEI hardware partners

Pilot Service Providers

Open Calls' solutions

External contributors (e.g., SMEs, projects)

IPCEI-CIS Reference Architecture



Key Concepts

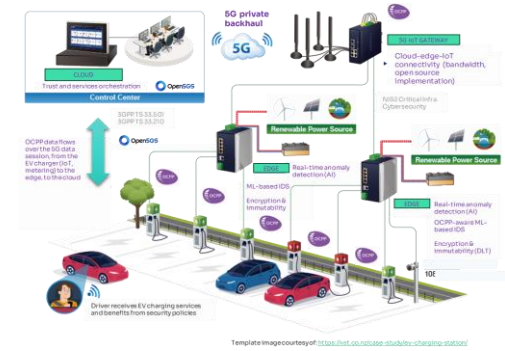
O-CEI orbits into certain key concepts:

- **Marketplace:** Offers digital assets that consumers can retrieve, including blueprint products, best practices and guidelines for their deployment.
- **Blueprints:** Intentful declarations of an operative target to be materialized in a CEI environment, delivered in a simple and concise language via short document.

BP9: Adaptive Cyber Monitoring and Trust Management of DERs via 5G-Enabled Distributed Operations

Target Declaration

This blueprint instructs on how to **secure mission-critical Distributed Energy Resources** (e.g., EV chargers, storage, PV) by integrating ultra-reliable, low-latency **5G connectivity with continuous cybersecurity assurance**. It leverages a Cloud-Edge-IoT architecture where AI at the edge performs real-time anomaly detection, while dynamic trust scoring ensures that only verified, tamper-free data influences grid operations; enforcing security and resiliency while ensuring latency



Technical Elements



- Bidirectional chargers and stationary storage integration via **secure IoT gateways/hubs**
- **5G Private network backbone**, ensuring deterministic low-latency and ultra-high reliability for critical control loops
- **Hybrid CECI architecture**, with edge hosting among detection models and cloud managing trust and services orchestration
- **Adaptive Cyber Monitoring stack**, with signature and ML-based IDS and anomaly detection to identify known and zero-day threats and emerging attack patterns
- **Trust & Integrity Layer**: end-to-end encryption, cryptographic integrity verification, and tamper-evident logging for auditability

Strategic Conditions and Decisions



- **Define 5G scope**, securing private spectrum or a slice to guarantee bandwidth and isolate critical traffic from public networks
- **Implement adaptive retraining loops** where the Cloud updates Edge AI models based on new threat data collected across the fleet
- **Ensure the system implement fails-safe mechanisms** if communication is lost
- **Establish security policies** where control commands are only executed if the source device's **trust score** meets a defined threshold

Standards and expert references

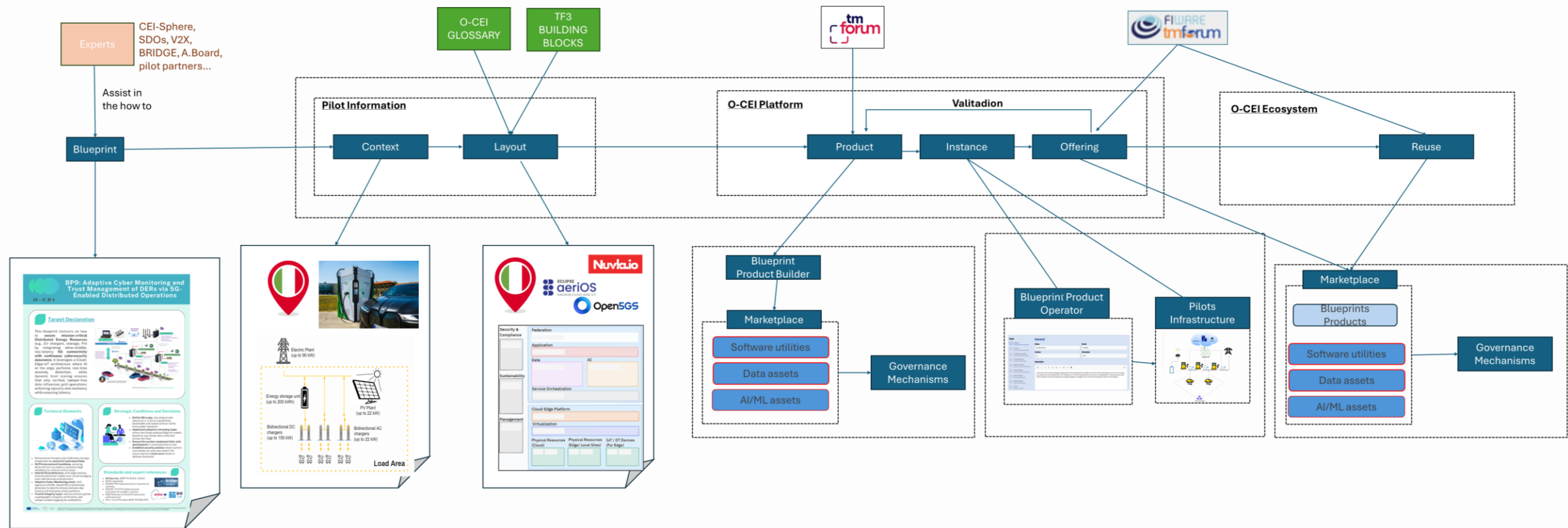
- **5G Security:** 3GPP TS 33.501, 33.210
- EUCS regulation
- ISO/IEC PWI Cybersecurity of systems of systems,
- ISO/IEC TS 27115 Cybersecurity evaluation of complex systems
- NIS2 Directive (critical infrastructure cybersecurity)
- Zero-Trust Principles (NIST SP 800-207)



Blueprint Lifecycle – Rollout process



The Blueprint lifecycle covers a range from design-time abstraction to run-time instance and Reuse.



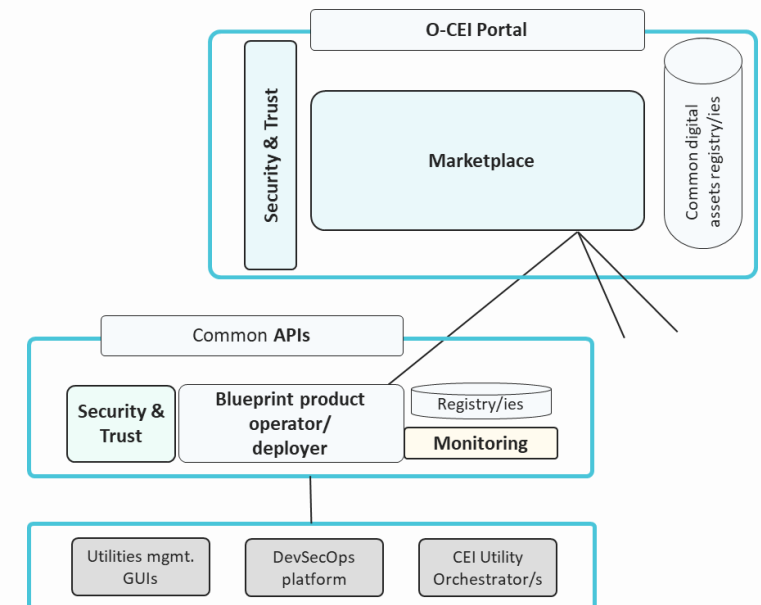
Key Concepts

O-CEI orbits into certain key **concepts**:

- **Marketplace:** Offers digital assets that consumers can retrieve, including blueprint products, best practices and guidelines for their deployment.
- **Blueprints:** Intentful declarations of an operative target to be materialized in a CEI environment, delivered in a simple and concise language via short document (full definition in O-CEI_D2.2).
- **O-CEI Platform:** Open-source solution that will guide adopters and automate the deployment of an (acquired) digital asset within a specific distributed computing ecosystem.

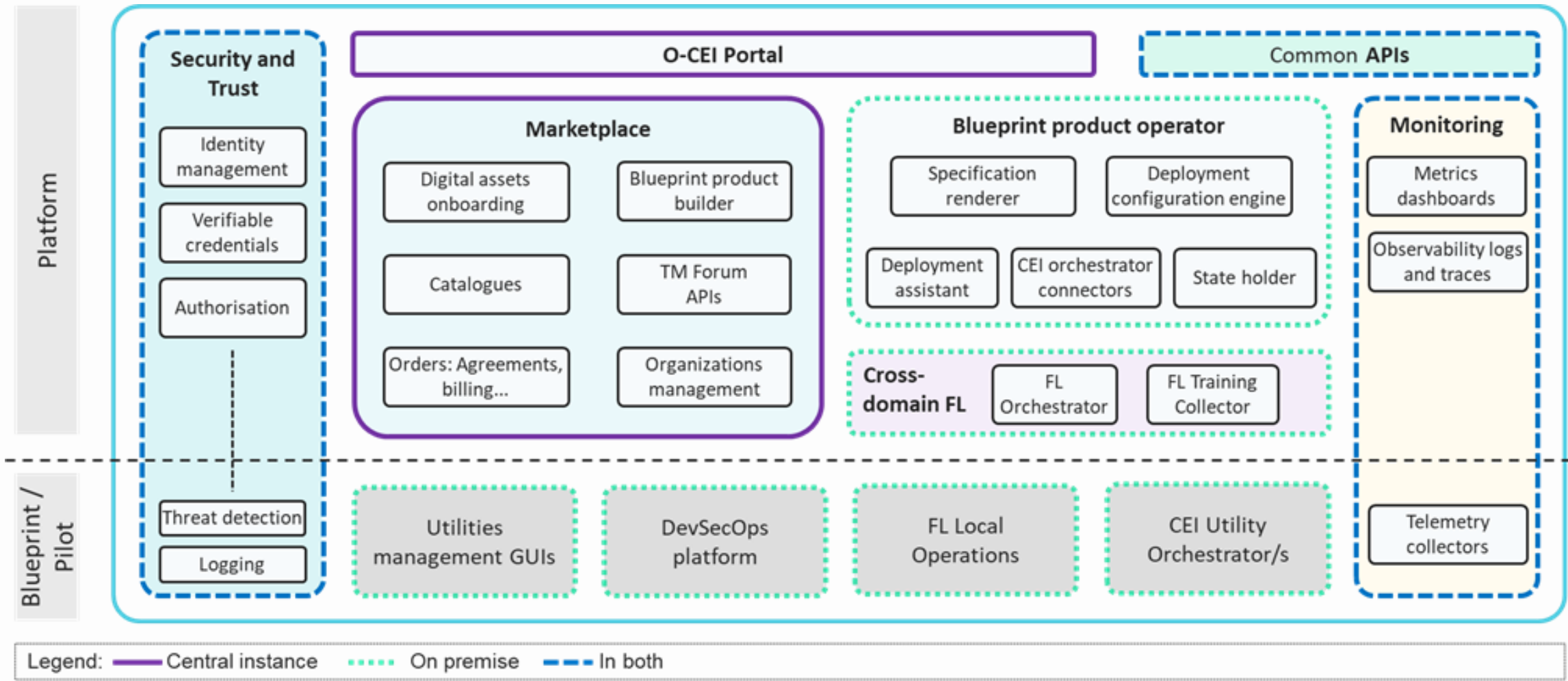
In charge of creating and deploying the blueprints and digital assets

Part central, Part in every pilot



O-CEI Platform Overview

- In cloud/central instance
- On premise
- In both



O-CEI Pilots towards OC#1

Global overview and constraints

Pilots in a Nutshell – Pilot 1

Energy Grid performance optimization upon RES integration →

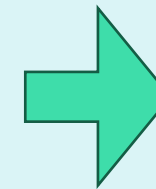


Scenarios

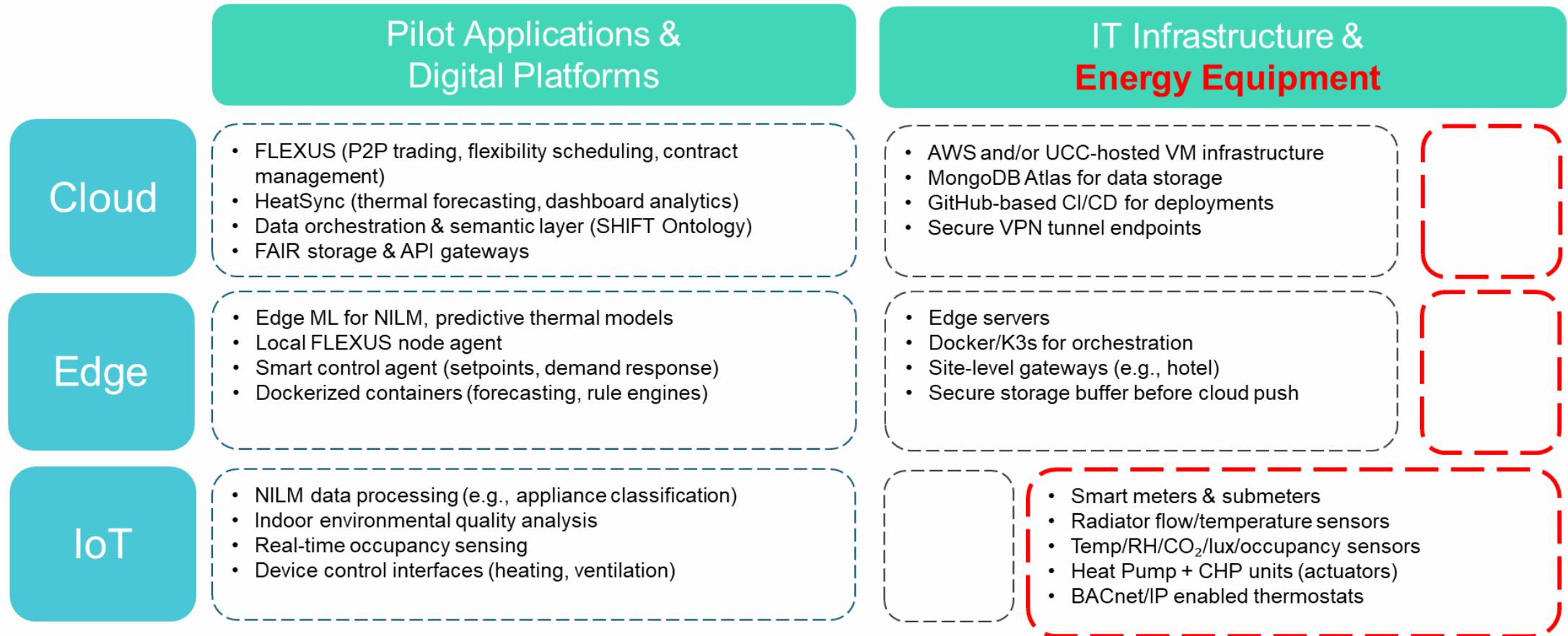
1. Community-Driven RES Integration by empowering prosumers (IR)
2. Smart thermal Load Management (IR)
3. Wide Smart grid Optimization (KR)
4. EV and Heat Pump Synchronization (KR)
5. CEI-Based energy management services for end-consumers (FR)
6. Virtualisation for smart substations to improve RES integration (FR)

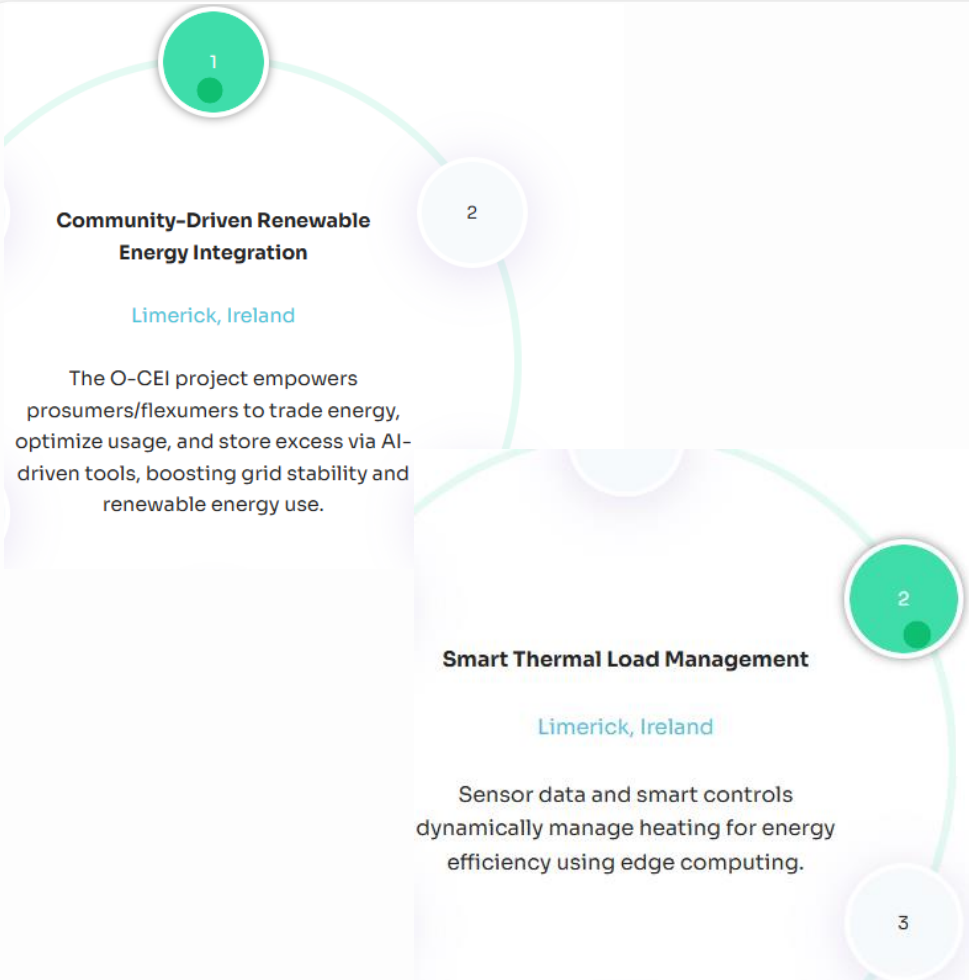
To improve **energy flexibility**, increase **renewable energy integration**, and reduce **carbon emissions** by leveraging **innovative technologies** and actively **engaging local communities**.

Reduce **Energy Consumption**
Improve **User Satisfaction**
Reduce **CO2 generation**
Increase **Use of RES**
Reduce **Peak Energy Demand**
Increase **Stability of Grid Voltage**



Consumer cost saving
Community engagement
More efficient RES integration
Better grid demand calculation and response





- Adjusted Load Forecaster (ALF) (Ireland scenario)

MSEMICON
ALF TEORANTA

ciaran.obreartuin@msemicon.com

- Mapping Flex Potential Across Cultures (Ireland scenario)

FlexPulseAI

Eduardo Ulloa

ulloa@novatechs.es

- Synthetic Energy Data Engine

SEDGE

Xilbi SI

pedro.branco@xilbi.com

Pilot 1.1 (IE) Expressed Concerns



- Standard IoT cybersecurity practices must be applied
- Edge deployment hardening must be validated during integration.
- Strong coordination is required.
- No major concerns.

Pilot Applications & Digital Platforms

IT Infrastructure & Energy Equipment

Cloud

- “DM4I” (Data Management For Innovation) digital platform for secure data storage and processing, on EDF R&D premises



- Databases
- CPU/GPU data processing environments & clusters

- IA-based algorithms for load curve and energy consumption analysis (e.g., disaggregation)

Edge

- Data gathering and local storage and processing
- Mobile application sandbox for testing of innovative energy services that make use of the processed data
- Data processing / AI deployment capabilities on the Gateway object
- Focus on Locality/Privacy

- Android/iOS mobile application



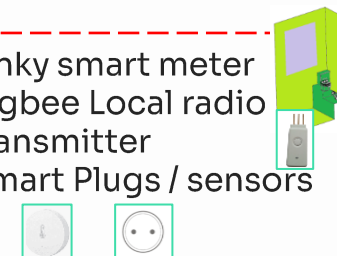
- Deployment of local load curve analysis algorithms

IoT

- Wifi/Zigbee Gateway for data gathering, storage and processing
- Zigbee mesh network



- Linky smart meter
- Zigbee Local radio transmitter
- Smart Plugs / sensors



Cloud

Pilot Applications & Digital Platforms

- Distribution level SCADA
- Dynamic voltage regulation application

IT Infrastructure & Energy Equipment

- Network & Substation Level control server
- Gateway

- Telecommunication infrastructure (optic fiber and wireless transmitter)

Edge

- Local voltage regulation application
- Grid protection functions
- Circuit breaker and Load-Tap changer control system

- IED (Intelligent Electronic Device) / IEC 61850 relay
- Router
- Switch

- MV/LV transformer
- On Load Tap changer
- Circuit breaker
- MV & LV measuring equipment

IoT

- Average measure computation application in real time (10 min)

- Linky Smart meter
- DEIE / DER management system, concentrators

- LV measuring equipment
- Renewable sources (PV, wind turbine)
- Consumption Load (residential)





- Innovative Edge-AI for Energy Management and Decision Making (France scenario)

	HyFHE-Net	Random Red	kruno@randomred.eu
ELEXIR		PLEGMA LABS	stelios@pleg.ma

- Virtualized voltage control for smart grids

- Efforts assessment must be done. It is not acceptable the need of development from our stakeholders so that your projects can be completed (ELEXIR, HyFHE-Net).
- On-prem only: cloud-side must run inside EDF internal cloud (no external infrastructure).
- Data sovereignty: data cannot leave the EDF perimeter (EDF Gateway and EDF cloud); EDF retains full data ownership (contractually explicit).
- AI models: no public sharing of models trained on pilot data; agree public metadata vs private trained artefacts and model IP.
- EDF IS constraints: integration is conditional on strict EDF Information System requirements (non-negotiable).
- Capacity and dependency: APIs must be developed to expose meter/sensor data and this requires dedicated EDF engineering time (main timing risk).
- Legal / ownership issues about data are raised about project ELEXIR. Please, check provided sheet.

P1.3 (Croatia) Assets and Infra

Cloud Layer

- ⑩ The pilot relies on a **self-managed data center** model using dedicated HP servers.
- ⑩ Cloud-like services such as **data aggregation**, **smart grid optimization**, and **prosumer dashboards** are hosted locally.
- ⑩ A **redundant server farm** ensures availability and fault tolerance.

Edge Layer

- ⑩ **Real-time processing** of data from smart meters and sensors.
- ⑩ Edge servers manage **grid balancing algorithms**, **load synchronization**, and **prosumer analytics**.
- ⑩ Hybrid **Inverters**, Local **Controllers**, Battery Management System (**BMS**) **modules**, and Household **gateways**

IoT Layer

- ⑩ Local Linux-Mynt-based nodes interface with **prosumer meters**, **PV generation systems**, **storage**, and **consumption tracking devices**.
- ⑩ **HEP** 15-min data feeds, **Modbus-connected** heat pumps, **Zigbee** smart plugs and sockets, sensors for household monitoring.



Wide Smart Grid Optimization

Krk Island, Croatia

Seasonal demand patterns are optimized with renewable integration and decentralized learning models.

3

EV and Heat Pump Synchronization

Krk Island, Croatia

Synchronizing renewable energy for household appliances and EVs ensures comfort and charging availability.

3

- Application of artificial intelligence technology (Croatia scenario)

Implementation of Blockchain technology in smart grids (Croatia scenario)

ENTRUST

Advanced Microturbines Srl

e.guglielmino@microturbines.it

Cybersecurity for Smart Grids at the Edge (Croatia scenario)

HYPEREDGE-SG

Cyentific AS

zych@cyentific.eu

- Care must be taken to ensure data protection and avoid misuse of energy transaction data at individual prosumer level.
- Latency and scalability under real microgrid conditions must be carefully validated.
- Further clarification on regulatory compliance is needed.
 - To be included in IMP (ENTRUST).
- Incidental findings must be protected.
- No major concerns expressed.

Software Defined Vehicle for VAS in Urban Areas →



Scenarios

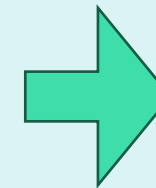
1. VaS for vehicle safety
2. VaS for sustainability
3. VaS for fleet V2G integration
4. VaS for grid optimization

The goal is to achieve a **sustainable Vehicle-as-Software (VaS) environment** at large scale in **urban areas**.

Reduce Non-exhaust **emissions**

Reduce **Reaction time** to incidents

Reduce **Energy consumption** of VaS Vehicles

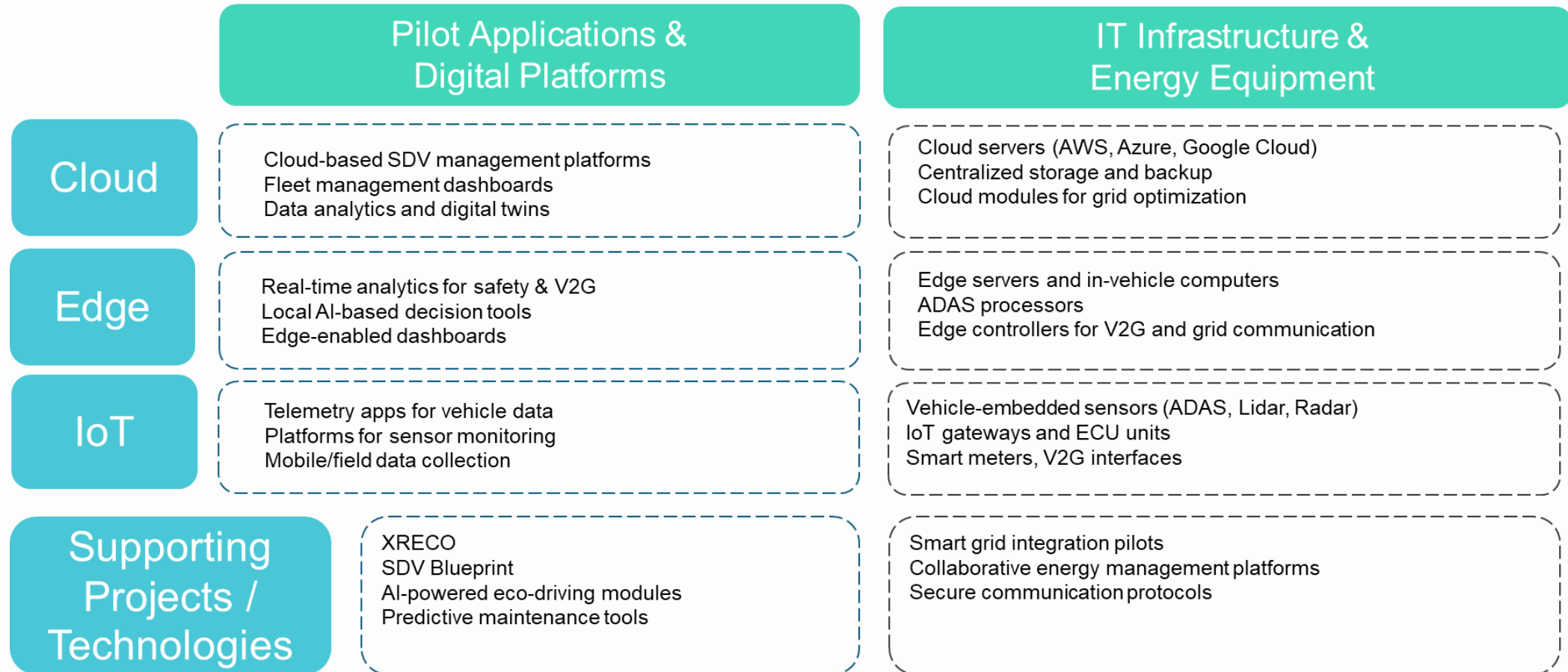


Reduction in emissions

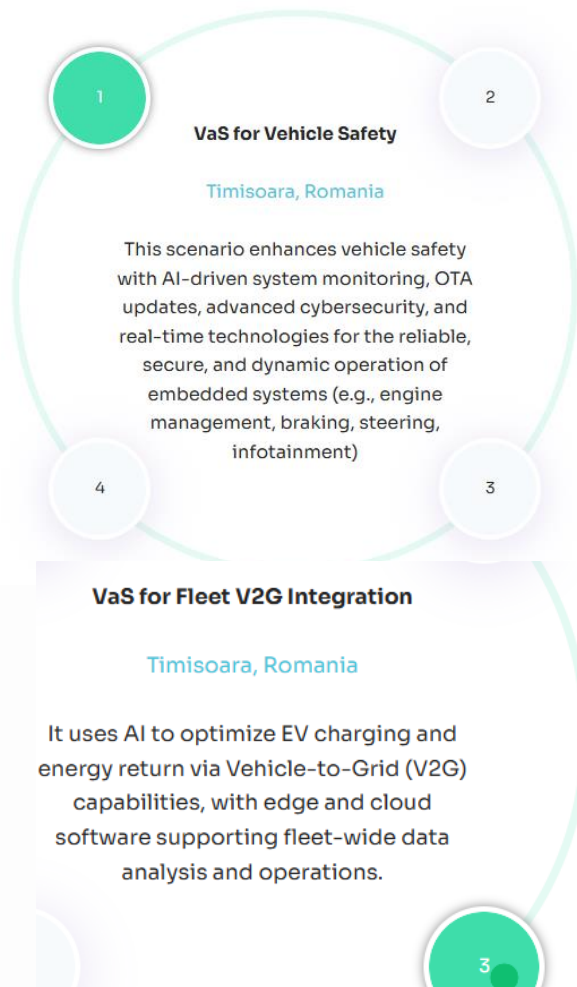
Faster response to critical events by VaS

Reduction of accident rates

P2 (Romania) assets and infra



Pilot 2 Challenges



- **Vehicle as Software (VaS) for Vehicle Safety**

SAFE-EDGE eProsima JaimeMartin@eProsima.com

- **Vehicle as Software (VaS) for Sustainability:**

EYE-ACCURATE Kentyou levent@kentyou.com

- **Vehicle as Software (VaS) for Fleet V2G Integration:**

SV2G-Twin HiveIntel GmbH danh@hiveintel.de

- **Vehicle as Software (VaS) for Grid Optimization**

- Security architectures in technology are not defined in detail and require further clarification in all 3 projects.
- Certain dockerised services may run only over test infrastructure (to be assessed case-by-case).
- On-site hardware availability must be assessed case-by-case.
- SAFE-EDGE: Depth of integration (especially around ECUs/controllers) must be clarified.
- EYE-ACCURATE: **Assumptions of easy access to vehicle signal access might be an issue** (this depends on OEM/IP). Applicant must be prepare an alternative.

Energy consumption and emission reductions in postal service fleet operation via intelligent BEV charging strategies →

Scenarios

1. Optimizing Storage Battery Systems
2. V2G as mobile storage unit
3. Smart and Innovative energy management systems



Fleet manager



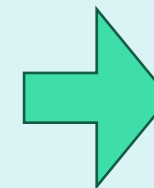
System
Integrator



RE chargers

Optimize the charging scheduling to take advantage of EV's flexibility as a load while **minimizing the impact of renewables** to the grid.

Reduce **total cost** of ownership and operations
Improve **Charging processes**
Reduce **Charging times**
Increase **Distance traveled per charge**
Reduce **Idle time**
Reduce CO2 Emissions



Fleet optimization
Deployment of Smart storage strategies
Enable a monitoring and adaption framework
Fleet vehicles as mobile energy storage

Pilot 3 (Austria) Assets and infra

Pilot Applications & Digital Platforms

IT Infrastructure & Energy Equipment

Cloud

Edge

IoT

- Energy procurement optimization [ENSTEI]
- Energy supply contracts [ENSTEI]
- Adapt energy storage system [POST]
- Real-Time Optimization of costs and revenues [POST]
- Optimization of Charging and Discharging Cycles [POST]
- Optimization Agent [AVL]



- Reinforcement learning framework
- Behavior model (for each subsystem)
- Third party services (e.g., weather)
- Battery capacity simulation

- Vehicle Connectivity (heterogenous)
- Infrastructure connectivity (measurement devices)



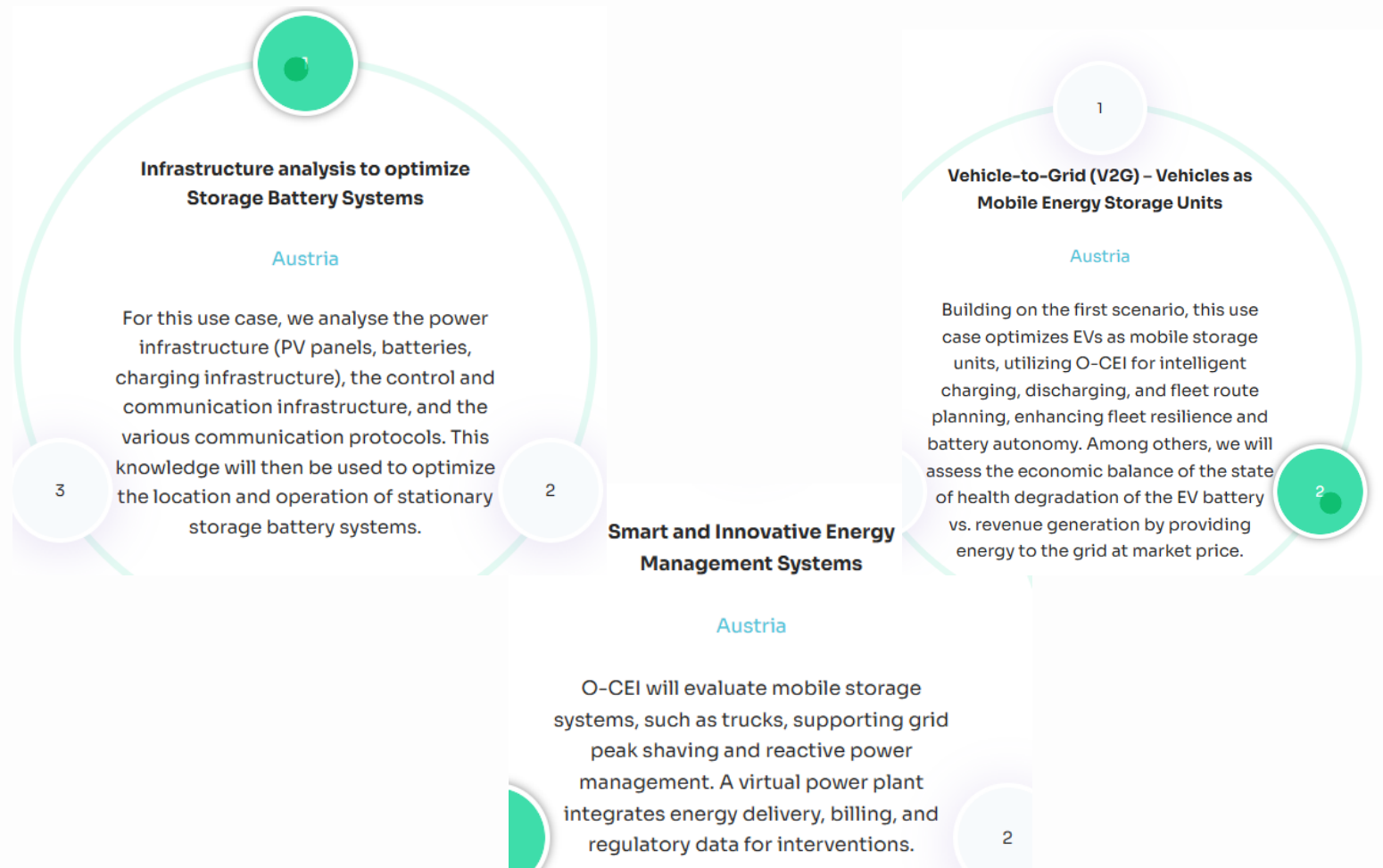
- Smart Energy Management System
- Historic energy market data
- Load profiles

- Energy market data
- PV generation data

- Live Data (vehicle)
- Bi-directional charging infrastructure
- PV plants
- Stationary battery storage



Pilot 3 OC Challenges



- **Services to ease the data sharing process:**

PrivacyGuard-Fleet InfAI Infinity GmbH funding@infai.org

- **Performing ML training process at the edge:**

ARIEL INDIGMA INNOVATIONS info@indigma.eu

- **Dealing with legacy assets (IT, energy):**

- **Synthetic data generation:**

Multiverse ignacio.torrens@multiversecomputing.com
SynthiaFleet Computing

- The post does not allow external software to be operated on the depot edge due to security reasons.
- The post does allow external software to be operated on the depot edge due to security reasons (GPS positions of drivers cannot be obtained).
- Not possible to connect to depot edge infrastructure during the OC (PrivacyGuard-Fleet, ARIEL). Check document and propose alternative (major concern).
- Completeness of historic data must be properly assessed.
- To be followed up in private.

THANKS

FOR YOUR ATTENTION

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Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research EAER
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